

Panel Data and the New Diff-in-Diff

CML Week 9

2026-11-03

Schedule

1. Design-based causal inference & Monte Carlo simulation
2. Covariate adjustment
3. Balancing weights
4. Double robustness, Double ML and TMLE
5. Heterogeneous Treatment Effects
6. HTE II: Neural Networks
7. Off-policy evaluation and optimization
8. Experimental design
9. **Panel data and the new diff-in-diff**
10. Partial Identification
11. Response adaptation
12. Interference



Readings for Today

Foundational

› (Abadie, Diamond, and Hainmueller 2010): Synthetic control methods for comparative case studies

Common

› (Roth et al. 2023): What's trending in difference-in-differences?

Presenter

› (Callaway and Sant'Anna 2021): Difference-in-differences with multiple time periods

› (Arkhangelsky et al. 2021): Synthetic difference-in-differences

From experiments to observational panel data

Last week: Randomization ensures treatment assignment is independent of potential outcomes
This week: Can we get identification without randomization?
The promise of panel data:

- Observe units before and after treatment
- Use pre-treatment trends to infer counterfactual post-treatment outcomes
- "Parallel trends" replaces randomization

The claim: If treatment and control groups would have followed parallel trends absent treatment, differencing removes time-invariant confounding
Spoiler: This is harder than it looks!

Difference-in-differences

The canonical difference-in-differences

Setup: Two groups (treatment/control), two periods (pre/post)

	Pre $(t = 0)$	Post $(t = 1)$
Control $(D = 0)$	$\bar{Y}_{00}$	$\bar{Y}_{01}$
Treatment $(D = 1)$	$\bar{Y}_{10}$	$\bar{Y}_{11}$

The DiD estimator:

$$\hat{\tau}_{DiD} = (\bar{Y}_{11} - \bar{Y}_{10}) - (\bar{Y}_{01} - \bar{Y}_{00})$$

Intuition:

- First difference: change in treatment group
- Second difference: subtract change in control group (the counterfactual trend)
- Result: treatment effect, purged of common time trends



Identification: What does parallel trends buy us?

The fundamental problem: We never observe $Y_{i1}(0)$ for treated units
Parallel trends assumption:

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0]$$

In words: *Absent treatment, treated and control groups would have experienced the same average change*
What this gives us:

$$\begin{aligned}\mathbb{E}[Y_{i1}(0) \mid D_i = 1] &= \mathbb{E}[Y_{i0}(0) \mid D_i = 1] + \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0] \\ &= (\text{observed}) + (\text{observed})\end{aligned}$$

The counterfactual is identified! We can estimate the ATT:

$$\tau_{ATT} = \mathbb{E}[Y_{i1}(1) - Y_{i1}(0) \mid D_i = 1]$$

The hardness of parallel trends

The core problem: Parallel trends is an assumption about *counterfactuals*

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0]$$

We never observe $Y_{i1}(0)$ for treated units $\implies$ **fundamentally untestable**

What we're really assuming:

- Treatment and control have the same unobserved time-varying factors
- Selection into treatment based only on:
  - Time-invariant differences (removed by differencing)
  - Observed time-varying factors (can condition on these)
- *No time-varying unobserved confounders that differ by group*

This is strong! Requires that all differential trends are due to treatment, not selection

Testing and falsification

Can we test parallel trends? Not directly, but we can check *pre-trends*
Pre-trends test: Do treatment and control have parallel trends *before* treatment?

- If yes: suggests parallel trends might hold post-treatment
- If no: parallel trends assumption is likely violated

The logic: If trends differ pre-treatment, why would they be parallel post-treatment?
Limitations:

- Pre-trends can pass even when parallel trends fails
  - Example: Firms change behavior *before* policy announcement, masking differential pre-trends
- Pre-trends can fail even when parallel trends holds
  - Example: Treatment targets units with temporarily low outcomes; pre-trends diverge but would have converged absent treatment
- Limited pre-treatment periods $\implies$ low statistical power
- Absence of evidence $\neq$ evidence of absence

Bottom line: Pre-trends are good but not determinative of parallel trends

When is parallel trends sensitive to functional form? (Roth and Sant'Anna 2023)

The insight: Parallel trends can hold in one scale but not another
Example: Consider earnings Y and log earnings $\log(Y)$

- Levels: $\mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid D = 1] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid D = 0]$
- Logs: $\mathbb{E}[\log Y_t(0) - \log Y_{t-1}(0) \mid D = 1] = \mathbb{E}[\log Y_t(0) - \log Y_{t-1}(0) \mid D = 0]$

These cannot both be true! (except in degenerate cases)
The problem:

- Parallel trends in levels $\implies$ constant additive differences
- Parallel trends in logs $\implies$ constant multiplicative differences
- Your choice of scale is an *untestable assumption*

Why randomization is different: With random treatment assignment, $D \perp (Y(0), Y(1))$ holds in *all* functional forms

- Independence is scale-invariant

When does DiD go wrong?

The setting: Multiple groups, staggered treatment adoption
Two-way fixed effects (TWFE) regression:

$$Y_{it} = \alpha_i + \lambda_t + \tau D_{it} + \epsilon_{it}$$

The promise: $\hat{\tau}$ estimates the average treatment effect
The problems:

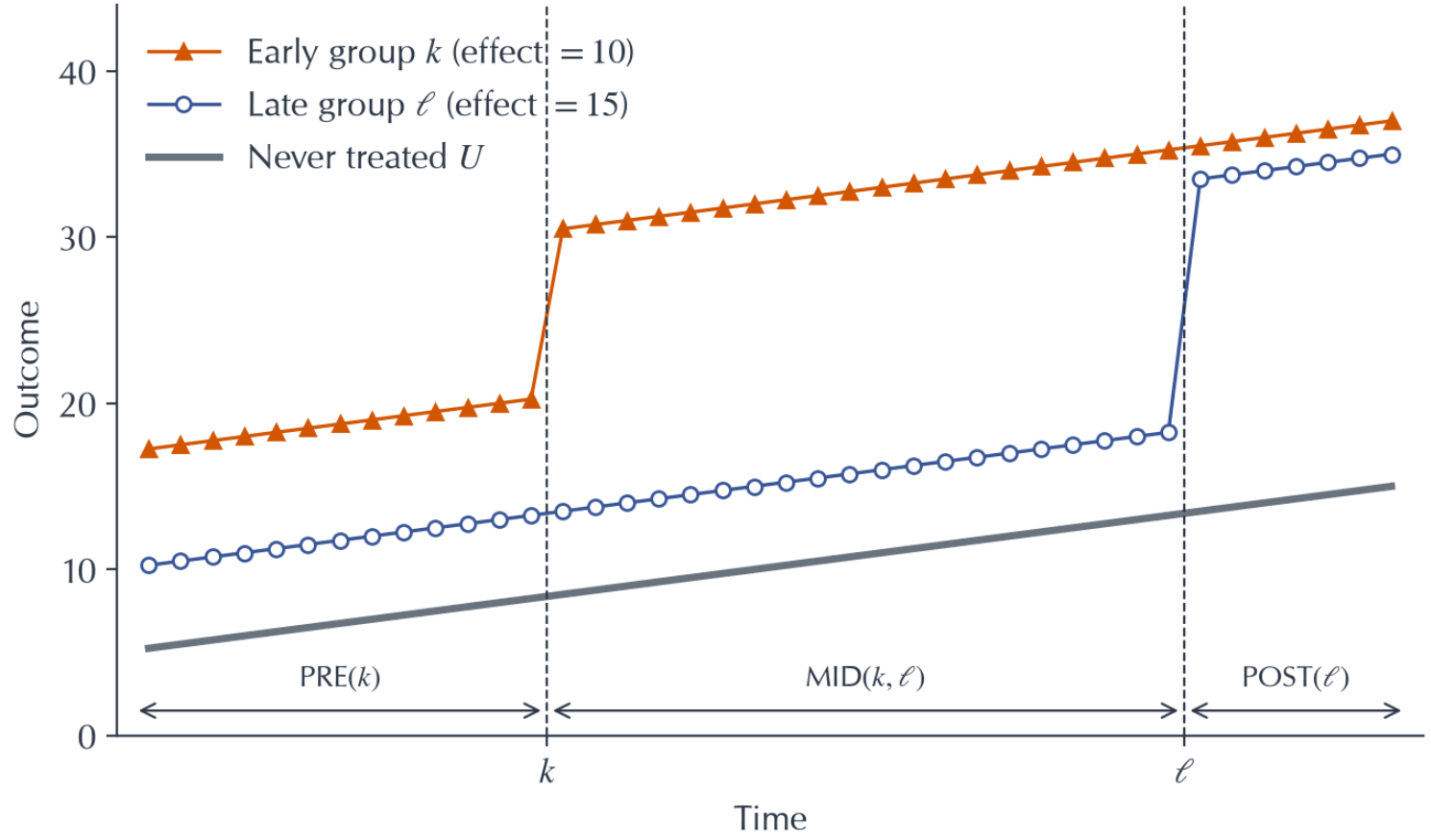
1. **Heterogeneous treatment effects:** If effects vary across groups/time, TWFE is a *weighted average* with potentially negative weights (De Chaisemartin and d'Haultfoeuille 2020)
2. **Forbidden comparisons:** TWFE compares already-treated units to newly-treated units, which violates parallel trends (Goodman-Bacon 2021)
3. **The result:** $\hat{\tau}$ can be biased, even with the sign flipped!

Bottom line: Basic TWFE with staggered adoption can be perilous

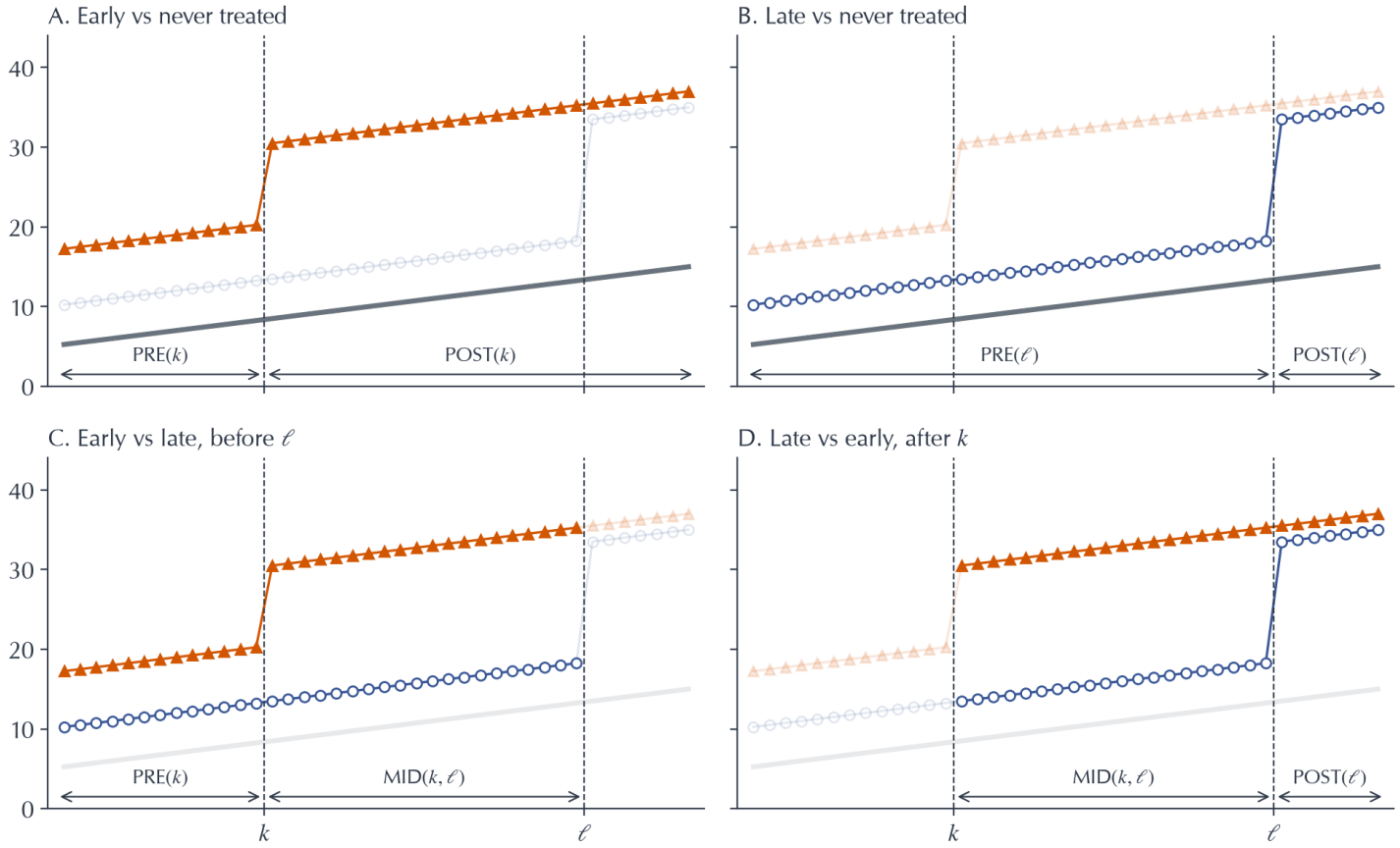
(Goodman-Bacon 2021)

Three timing groups

Early group k, late group ℓ, never-treated group U. Effects are constant: 10 for k, 15 for ℓ.



TWFE is an average of 2x2s



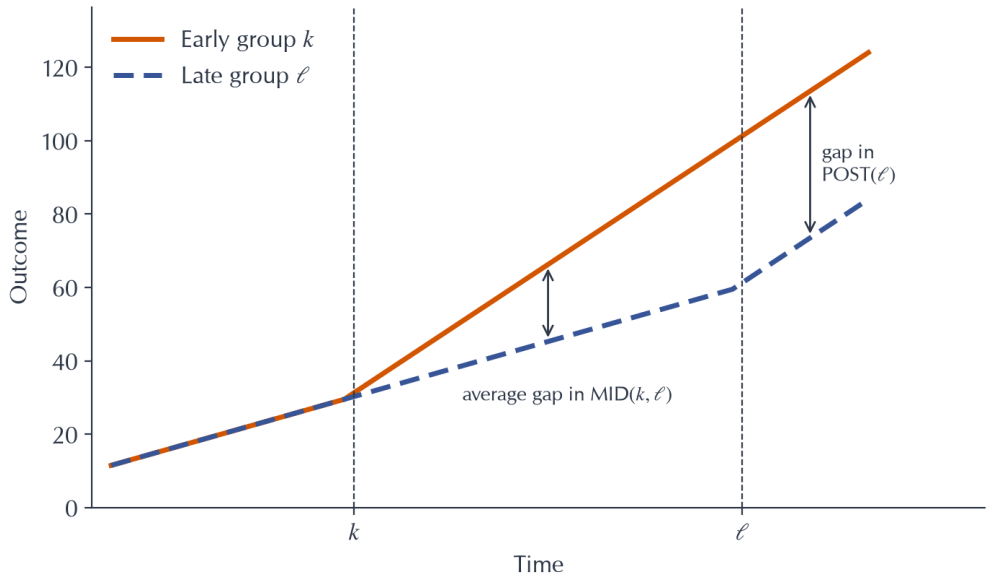
Theorem: $\hat{\tau}_{TWFE} = \sum s \cdot \hat{\beta}^{2\times2}$ over all four comparisons, with weights $s \geq 0$, $\sum s = 1$, set by group sizes and the variance of D_{it} in each window

The forbidden comparison

Panel D uses the **already-treated** group k as the control for ℓ. Under parallel trends:

$$\hat{\beta}^{2\times2,\ell}_{k\ell} \to \mathrm{ATT}_\ell(\mathrm{POST}) - \left[\mathrm{ATT}_k(\mathrm{POST}) - \mathrm{ATT}_k(\mathrm{MID})\right]$$

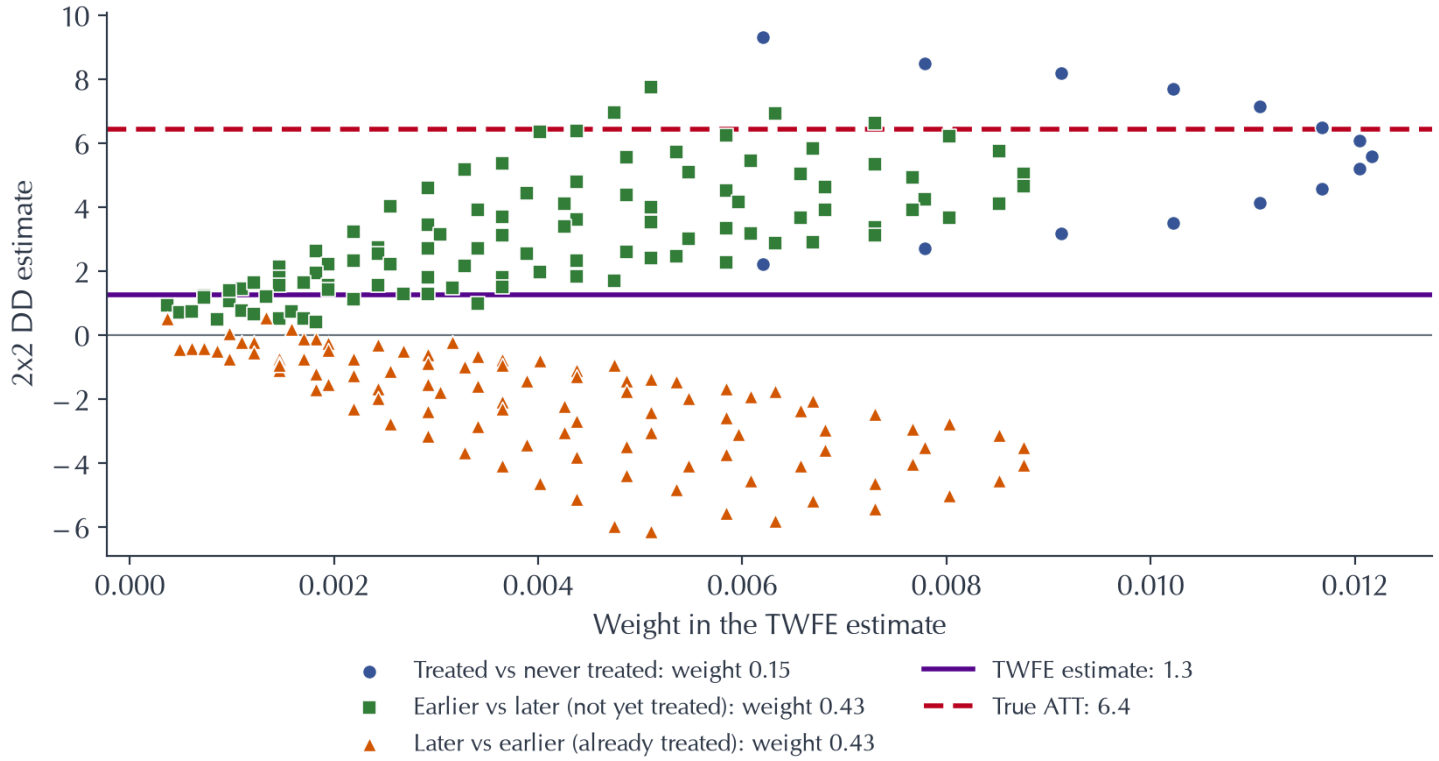
- Constant effects: the bracket is zero, and panel D is fine
- Effects that grow with exposure: the growth in k's effect is subtracted from ℓ's



- Identical counterfactuals; the effect grows linearly from adoption
- In POST(ℓ) both groups grow equally fast, so the gap stops widening
- Panel D sees the gap *shrink* relative to MID: an effect ≤ 0

The decomposition, simulated

Parallel trends holds. Every unit's effect is positive and grows by 1 per period of exposure.



Every weight is positive; the **later vs earlier** 2x2s go negative. This is where the negative weights of (De Chaisemartin and d'Haultfoeuille 2020) come from

(Callaway and Sant'Anna 2021)

Modern solutions

The key insight: Avoid forbidden comparisons by using never-treated units as controls (Callaway and Sant'Anna 2021; Sun and Abraham 2021)
Group-time ATTs: Estimate treatment effects for each cohort g at each time t:

$$ATT(g,t) = \mathbb{E}[Y_{it}(1) - Y_{it}(0) \mid G_i = g]$$

where G_i is the period when unit i first receives treatment
Aggregation: Combine group-time ATTs to get target parameters:

- Overall ATT: weighted average across all (g,t)
- Dynamic effects: average by event time $t-g$
- Group-specific effects: average over time for each g

Advantages: Valid inference, handles heterogeneity, avoids negative weights

Identifying $ATT(g, t)$

No anticipation: $Y_{it} = Y_{it}(0)$ for all $t < G_i$
Parallel trends, cohort by cohort: with $C_i = 1$ for never-treated units,

$$\mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid G = g] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid C = 1]$$

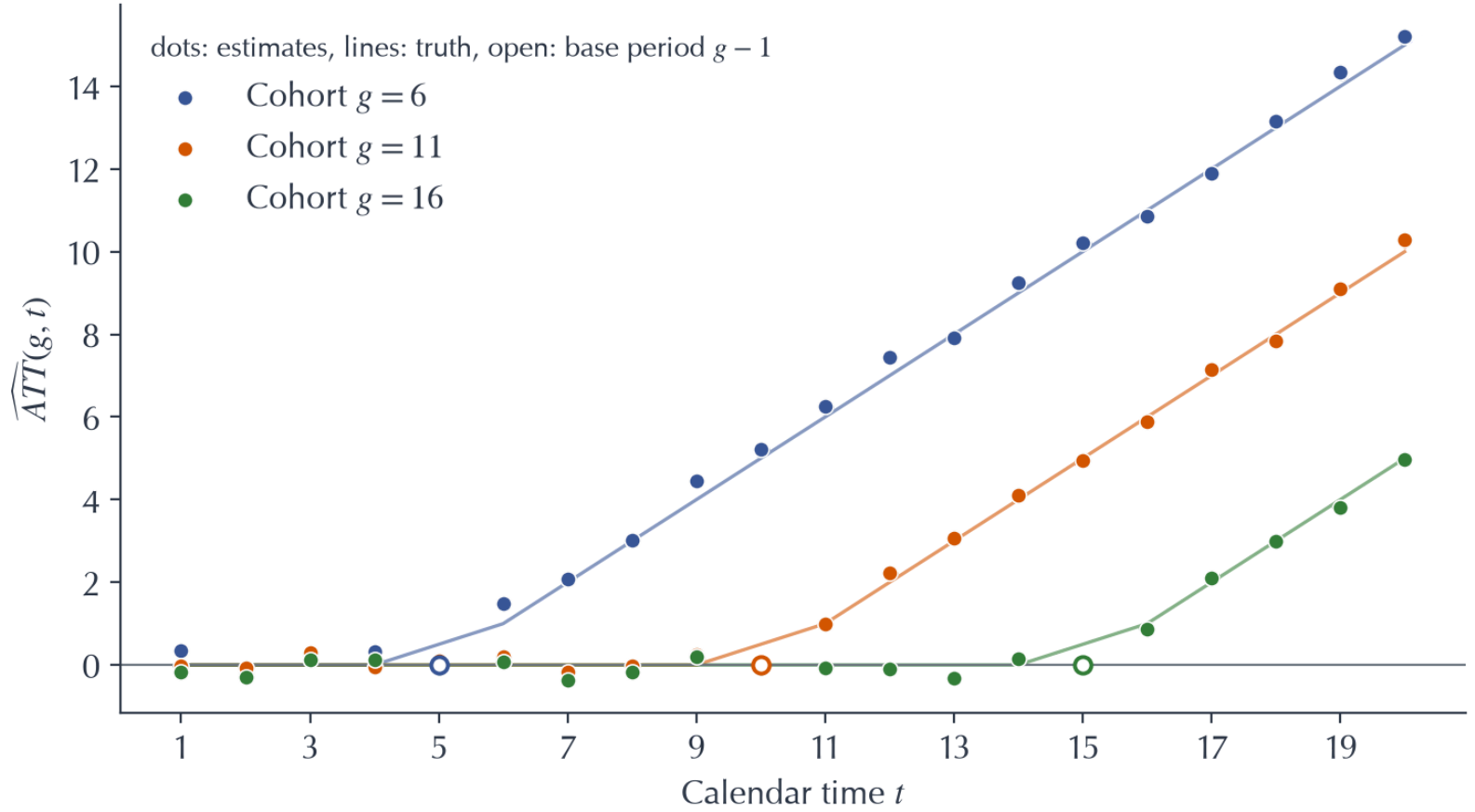
Then each $ATT(g, t)$ is one clean 2x2, with base period $g - 1$:

$$ATT(g, t) = \mathbb{E}[Y_t - Y_{g-1} \mid G = g] - \mathbb{E}[Y_t - Y_{g-1} \mid C = 1]$$

- Controls are **never treated**: no already-treated unit is ever a control
- Alternative: use the **not yet treated** ($G_i > t$) as controls, a larger comparison group
- With covariates, each 2x2 can be estimated by outcome regression, IPW, or the doubly robust estimator from Week 4

Group-time ATTs in a picture

Same kind of growing effect, three cohorts plus a never-treated group



› Points before $g-1$ are placebo estimates: a pre-trends check, cohort by cohort

Aggregating group-time ATTs

The **analyst chooses** the weights, and they are never negative

By event time (an event study), over the cohorts observed e periods after adoption:

$$\theta(e) = \sum_{g} \Pr(G = g \mid g + e \leq T)\, ATT(g, g + e)$$

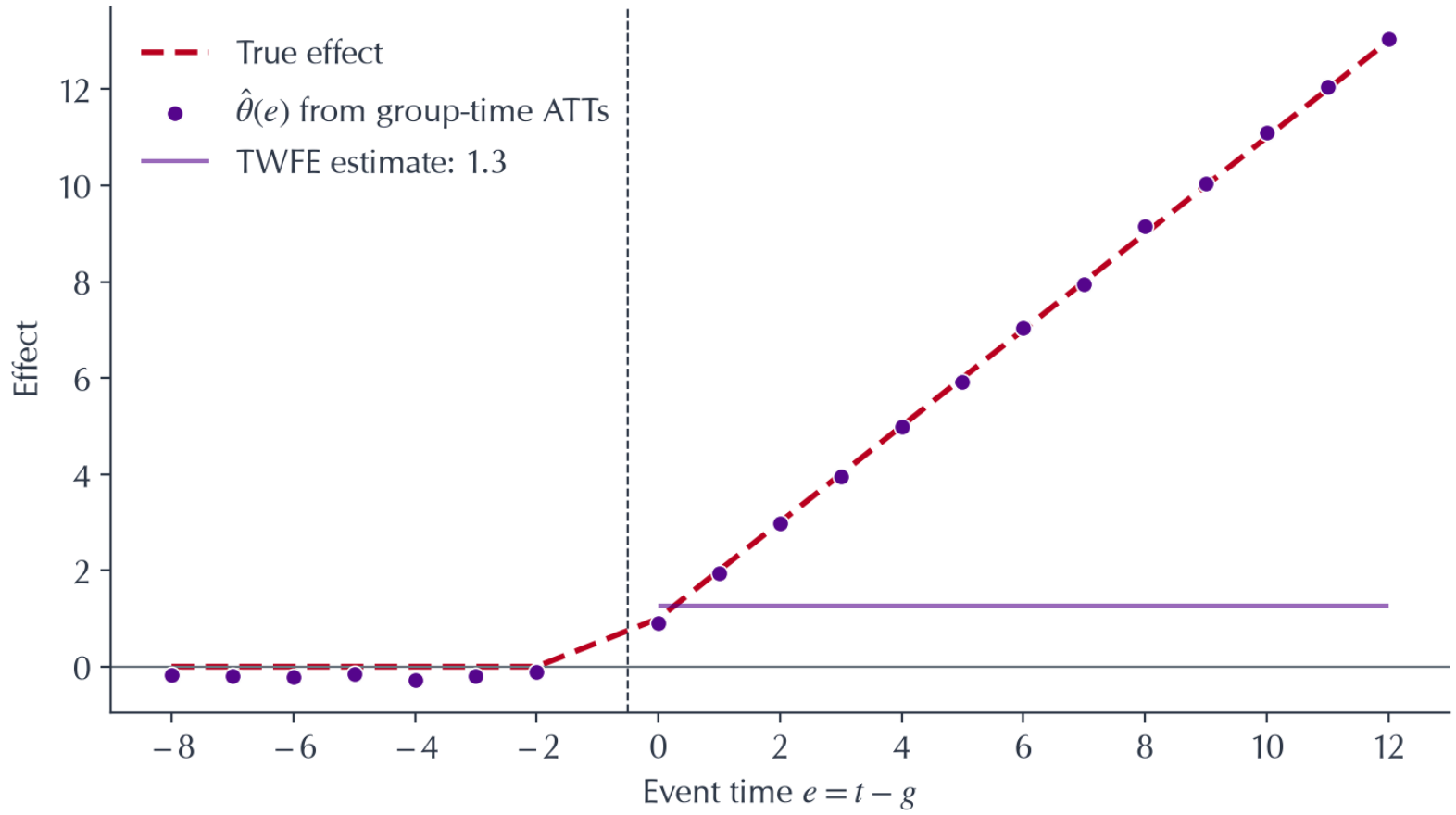
By cohort:

$$\theta(g) = \frac{1}{T - g + 1} \sum_{t \geq g} ATT(g, t)$$

Overall: average $\theta(g)$ with weights $\Pr(G = g \mid G \leq T)$

› Long horizons e come from early cohorts only: the mix of cohorts changes with e

Event study: group-time ATTs vs TWFE



Same simulated panel as the decomposition: true ATT 6.4, TWFE 1.3

(Abadie, Diamond, and Hainmueller 2010)

Synthetic controls: Focusing on parallel trends

The motivation: What if parallel trends doesn't hold for *any* single control unit?
Classic example: Effect of California's tobacco control program (Prop 99, 1988) (Abadie, Diamond, and Hainmueller 2010)

- One treated unit (California)
- Many potential control states (almost 50, in fact)
- But no single state is a good match for California

The insight: Create a *synthetic California* from other states

The synthetic control method

Setup: One treated unit, J control units (the "donor pool")
Goal: Find weights $w_j \geq 0$ with $\sum_j w_j = 1$ such that:

$$\sum_{j=2}^{J+1} w_j Y_{jt} \approx Y_{1t} \quad \text{for all pre-treatment periods } t < T_0$$

How to find weights: Solve optimization problem

$$\min_w \sum_{t<T_0} \left(Y_{1t} - \sum_j w_j Y_{jt} \right)^2 \quad \text{subject to } w_j \geq 0, \sum_j w_j = 1$$

- Why $w_j \geq 0$?
- Why $\sum_j w_j = 1$?

Identification: If it matches trends pre-treatment, maybe also post-treatment?

Why synthetic control works

Key assumption: The outcome follows a factor model

$$Y_{it} = \delta_t + \theta_t Z_i + \lambda_t \mu_i + \epsilon_{it}$$

where δ_t are time effects, Z_i are observed covariates, μ_i are unobserved factors

If weights reproduce:

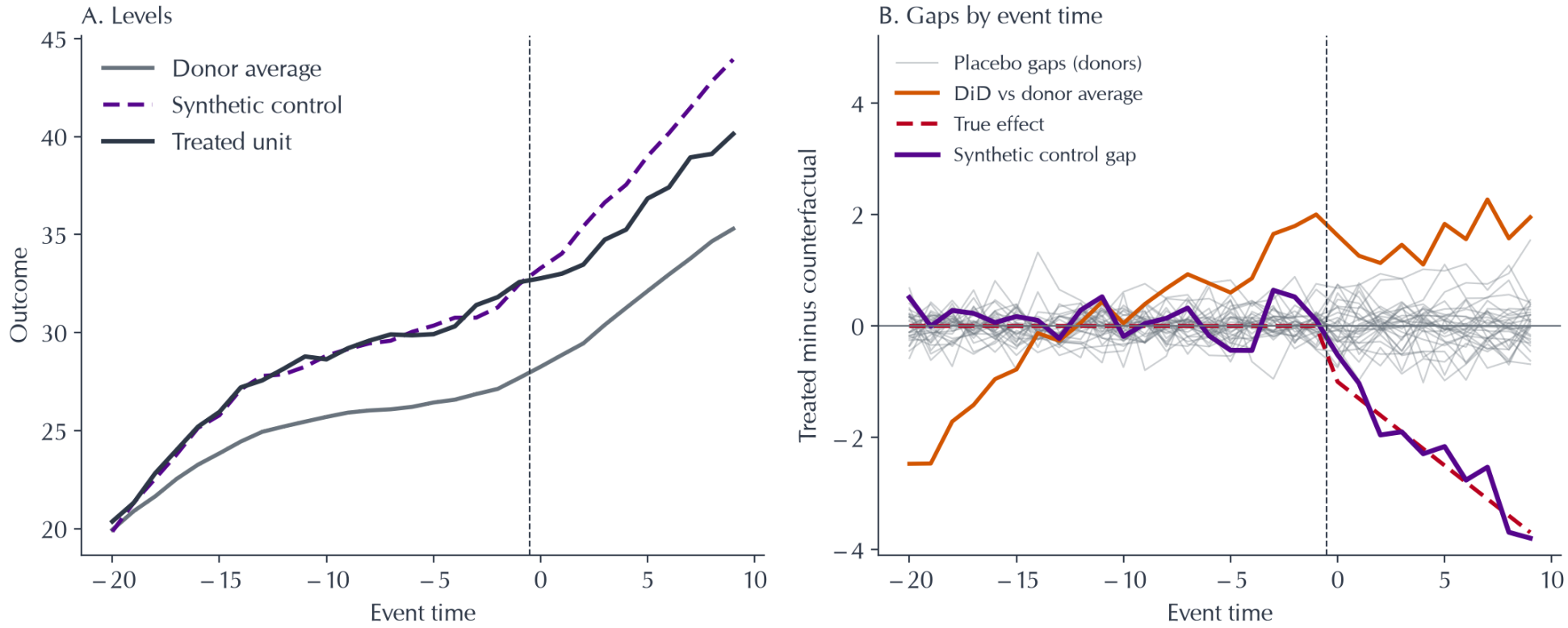
- Pre-treatment outcomes
- Pre-treatment covariates
- And we have enough pre-treatment periods

Then: The weighted control units have the same μ_i as the treated unit

Result: $\sum_j w_j^* Y_{jt}$ is an unbiased estimate of $Y_{1t}(0)$ post-treatment

Extensions: Augmented SC (Ben-Michael, Feller, and Rothstein 2021) adds outcome modeling for robustness

Synthetic control as an event study



- **Before:** the SC gap sits in the placebo band; DiD shows a **pre-trend**
- **After:** the SC gap tracks the true effect; DiD gets the sign wrong
- **Inference:** is the treated gap extreme among the donor placebos?

Synthesis

The evolution of diff-in-diff methods

Classic DiD (pre-2000s): 2x2 design with parallel trends assumption

› Simple and intuitive, but limited to two groups and two periods

TWFE era (2000s-2010s): Panel data with unit and time fixed effects

› Became the workhorse of applied econometrics

The reckoning (2018-2021): Discovery of fundamental problems

› Forbidden comparisons, negative weights, functional form sensitivity

Modern solutions (2020s): New estimation strategies

› Group-time aggregation, synthetic controls, sensitivity analysis

The lesson: Even “simple” methods hide complex assumptions

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